

How gamma changes access to the target under GD and Adam

Small tanh slopes can leave needed target components difficult to acquire under both gradient descent and Adam. On the same finite dictionary, the explicit spectral comparison below certifies rank-32 eigenvalue-ratio increases of approximately 35, 428, and 36,790 times relative to the actual gamma-8 ratio when gamma rises to 12, 16, and 64. At gamma 8, the chosen slow subspace initially holds 4.33% of the target energy. After 200,000 common-recipe Adam updates, its residual energy alone still exceeds the 1% error tolerance. At larger gammas it is below tolerance.

The distinction between rates and optimizer behavior remains essential. The GD spectrum predicts its training curve exactly, up to numerical evaluation. Adam changes the update geometry, so we measure residual projections. Its sustained crossings at gammas 12, 16, and 64 all occur around 41,000 updates under the common decaying-rate recipe. The spectral mechanism explains an observed small-gamma obstruction, but supplies neither an Adam time law nor a monotone ordering of sustained Adam times.

Symbol	Meaning
γ, h	Common tanh slope and center spacing
$J_\gamma, K_\gamma = J_\gamma J_\gamma^T$	Sample-normalized feature matrix and readout kernel
μ_i, u_i	Descending eigenvalues and orthonormal sample-space eigenvectors
$\rho_i = \mu_i / \mu_1$	Learning rate relative to the fastest direction under normalized-step GD
y, r_n	Normalized target samples and residual after n updates
$p_i = u_i^T y ^2 / \ y\ ^2$	Fraction of initial target energy in direction u_i
$P_\gamma(\rho), R_{\gamma,n}(\rho)$	Initial and remaining energy in positive modes with ratios at most ρ
H, T_{sust}	Adam horizon and first update below tolerance for its remainder

1 Why the largest eigenvalue matters

For inputs x_a , centers c_j , and m samples, use ordinary tanh features and a bias:

$$J_\gamma[a, j] = \frac{\tanh(\gamma(x_a - c_j))}{\sqrt{m}}, \quad J_\gamma[a, 0] = \frac{1}{\sqrt{m}}.$$

With normalized targets $y_a = f(x_a) / \sqrt{m}$, minimizing $\frac{1}{2} \|J_\gamma w - y\|^2$ is ordinary half mean-squared error. Geometry is frozen; only readout w is trained. From $w_0 = 0$, GD gives $r_{n+1} = (I - \eta K_\gamma) r_n$. The largest eigenvalue limits the stable step shared by all directions. For $\eta = 1/(2\mu_1)$,

$$\frac{\|r_n\|^2}{\|y\|^2} = \sum_i p_i(\gamma) \left(1 - \frac{\rho_i(\gamma)}{2}\right)^{2n}.$$

A ratio of one halves that component's residual every update. A ratio of 10^{-6} requires about two million updates for one e-fold reduction. A small ratio matters only if the target needs that direction. Both eigenvectors and target weights can change with gamma; ordered rank is not a fixed physical direction across dictionaries. For a prescribed common step, the rate is $\eta\mu_i$, and ratios alone do not determine update counts. The calculations below use the actual archived steps.

2 A finite-dictionary theorem with gamma explicit

This is the cross-gamma comparison from the collaborator's [alternative-view note](#), written with μ for eigenvalues. Centers are consecutive and uniformly spaced; inputs are fixed. Uniform input sampling makes the explicit increment Toeplitz and easy to apply, but the quadratic-form identity itself does not require uniform samples.

Let there be W centers, spacing h , and center-cell interval $I = [c_1 - h/2, c_W + h/2]$. Define $d_{ab} = x_a - x_b$ and

$$K_\gamma^{(0)}[a, b] = \frac{W+1}{m} - \frac{2}{hm} d_{ab} \coth(\gamma d_{ab}), \quad E_\gamma = K_\gamma - K_\gamma^{(0)},$$

using the continuous diagonal value $d \coth(\gamma d) = 1/\gamma$ at $d = 0$. The correction E_γ includes the finite center interval and replacement of a center integral by discrete centers. It is retained. For $\gamma_b > \gamma_a$, set

$$G_{ba}[a, b] = \frac{2}{hm} [d_{ab} \coth(\gamma_a d_{ab}) - d_{ab} \coth(\gamma_b d_{ab})].$$

Theorem 1 (Explicit gamma gain and finite spectral comparison). The matrix G_{ba} is positive semidefinite. For every real sample vector v ,

$$v^T G_{ba} v = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2} \left| \sum_a v_a e^{-i\omega x_a} \right|^2 d\omega,$$

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}.$$

Let U_i contain orthonormal eigenvectors for the largest i positive eigenvalues of K_{γ_a} . Set

$$C_i = \text{diag}(\mu_1(\gamma_a), \dots, \mu_i(\gamma_a)) + U_i^T G_{ba} U_i,$$

$$B_i = U_i^T (E_{\gamma_b} - E_{\gamma_a}) U_i, \quad \delta_i = \|C_i^{-1/2} B_i C_i^{-1/2}\|_2.$$

For any $L_b \geq \mu_1(\gamma_b)$,

$$\boxed{\frac{\mu_i(\gamma_b)}{\mu_1(\gamma_b)} \geq \frac{[1 - \delta_i]_+ + \mu_{\min}(C_i)}{L_b}}.$$

Here $\mu_{\min}(C_i)$ is the smallest eigenvalue of the displayed small matrix. The computed bound uses the maximum absolute row sum of the actual finite kernel for L_b .

The multiplier increases with gamma: larger gamma removes less high-frequency content. The theorem carries that gain into an entire trial subspace, then accounts for the finite correction and normalization by the largest eigenvalue. It does not assert that every finite-dictionary ratio increases. An improvement is guaranteed only when the bound exceeds the old ratio. The calculation uses an old eigenspace and small spectral problems; it is not a scalar law depending on gamma alone.

Proof

Write $g_{ab}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c))$. Direct integration gives

$$\int_{\mathbb{R}} (1 - g_{ab}(c)) dc = 2d_{ab} \coth(\gamma d_{ab}).$$

For distinct inputs, the identity $1 - \tanh u \tanh v = \coth(u-v)(\tanh u - \tanh v)$ reduces the integral to two translated tanh profiles. Their difference integrates to twice the translation. The coincident limit follows from $\int \text{sech}^2(\gamma c) dc = 2/\gamma$.

The center integral over I equals Wh minus the whole-line deficit plus the deficit outside I . Its normalized kernel is therefore $K_\gamma^{(0)}$ plus the omitted-exterior correction. Adding the discrete-center quadrature difference gives exactly $K_\gamma = K_\gamma^{(0)} + E_\gamma$.

The density $s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x)$ satisfies $\tanh(\gamma \cdot) = s_\gamma * \text{sign}$: both derivatives equal $2s_\gamma$ and both vanish at zero. Under convention $\widehat{f}(\omega) = \int f(x)e^{-i\omega x} dx$, substituting $u = e^{2\gamma x}$ and $a = \omega/(2\gamma)$ gives

$$\begin{aligned}\widehat{s}_\gamma(\omega) &= \int_0^\infty \frac{u^{-ia}}{(1+u)^2} du = \Gamma(1-ia)\Gamma(1+ia) \\ &= \frac{\pi a}{\sinh(\pi a)} = M_\gamma(\omega).\end{aligned}$$

This uses the beta integral and gamma reflection identity, with continuous value one at zero. The difference

$$D(d) = 2[d \coth(\gamma_a d) - d \coth(\gamma_b d)]$$

is integrable. The second derivative of $2d \coth(\gamma d)$ is $4(s_\gamma * s_\gamma)(d)$, by differentiating the deficit integral and integrating once by parts. Hence $\widehat{D}'' = 4(M_{\gamma_a}^2 - M_{\gamma_b}^2)$ and

$$\widehat{D}(\omega) = \frac{4(M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2)}{\omega^2}.$$

The continuous zero-frequency limit is $\widehat{D}(0) = \frac{\pi^2}{3}(\gamma_a^{-2} - \gamma_b^{-2})$. Fourier inversion contributes $1/(2\pi)$, giving precisely $2/(\pi h m)$ in the quadratic-form identity.

The larger-gamma multiplier is at least as large, so the transform is nonnegative. Fourier inversion proves the identity and positivity. In particular C_i is positive definite. Compressing the actual new kernel gives $U_i^T K_{\gamma_b} U_i = C_i + B_i$. By definition of δ_i , this is at least $(1 - \delta_i)C_i$ in quadratic-form order. Min-max yields $\mu_i(\gamma_b) \geq (1 - \delta_i)\mu_{\min}(C_i)$ for $\delta_i < 1$. When $\delta_i \geq 1$, positivity of the actual kernel gives zero as a lower bound. Divide by $\mu_1(\gamma_b) \leq L_b$. $\square$

3 From needed directions to a GD delay

Define positive slow energy

$$P_\gamma(\rho) = \sum_{0 < \rho_i(\gamma) \leq \rho} p_i(\gamma).$$

If $P_\gamma(\rho) \geq p > \varepsilon^2$, retaining these terms in the exact GD residual gives

$$E_\gamma(n) \geq \sqrt{p}(1 - \rho/2)^n, \quad n \geq \left\lceil \frac{\log(\sqrt{p}/\varepsilon)}{-\log(1 - \rho/2)} \right\rceil$$

as necessary time to reach relative error ε . For the actual archived step, replace $\rho/2$ by the per-update rate cutoff. The positive restriction separates slow acquisition from permanent nullspace error. The compact-filter calculation in the [structured-kernel note](#) supplies lower bounds on positive slow energy with its declared numerical allowances.

This time bound is a separate target-dependent statement. A lower bound on an eigenvalue ratio establishes a rate floor; it does not by itself establish a necessary delay. The full target-weighted spectrum gives the reference forecast; compact positive-energy bounds give conservative necessary times.

4 What we measure for Adam

[Adam](#) rescales readout-coordinate updates and uses gradient history. For $r_n = Jw_n - y$, its update is

$$r_{n+1} = r_n - \alpha_n J D_n \widehat{m}_n, \quad D_n = \text{diag}((\sqrt{\widehat{v}_n} + \epsilon_{\text{Adam}})^{-1}).$$

Raw-kernel eigenmodes are diagnostic coordinates, not independent Adam decay modes. Even without momentum, the effective kernel would be the time-dependent $J D_n J^T$. We do not apply the GD time law to Adam.

Instead, at saved checkpoints measure

$$R_{\gamma,n}(\rho) = \sum_{0 < \rho_i(\gamma) \leq \rho} \frac{|u_i(\gamma)^T r_n|^2}{\|y\|^2}.$$

Initially this equals $P_\gamma(\rho)$. At a fixed update count, it asks whether Adam removed target components occupying the GD-slow subspace. A substantial remaining value supports the proposed obstruction for this optimizer and protocol. A small value indicates acquisition of those components; that outcome must also be reported.

Sustained acquisition. With $H = 200,000$ and $\varepsilon = 0.01$,

$$T_{\text{sust}} = \min\{n : E(k) \leq \varepsilon \text{ for all } k = n, \dots, H\}.$$

If $E(H) > \varepsilon$, acquisition is unobserved within the horizon. For a successful run, this is one plus the last update above threshold, or zero when no update is above threshold. Report the confirmation length $H - T_{\text{sust}} + 1$. No smoothing is used and nothing is asserted beyond H .

5 Matched experimental protocol

The primary target is $f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$ on $[-1, 1]$. The archive uses spacing $h = 1/256$, 559 tanh features plus bias, and 8,193 uniformly spaced training inputs. Centers, samples, target, raw readout coordinates, and zero initialization are fixed; slopes are 8, 12, 16, and 64.

Common-recipe Adam uses initial rate 10^{-3} , additive epsilon 10^{-12} , and moments (0.9, 0.999). Its rate is constant through 20,000 updates, decays by cosine to 10^{-3} of its initial value at 50,000, then remains fixed. Runs continue to 200,000 updates. This schedule is a material part of the acquisition result.

Sensitivity comparisons reuse validation-selected recipes: five initial rates crossed with two epsilons, selected by median validation error at 40,000, 42,500, 45,000, 47,500, and 50,000 updates; optimizer states continue intact. No new selection uses present spectral projections. All plots concern training-grid optimization; separate validation data selected the sensitivity recipes. No new held-out generalization claim is made.

Supporting targets are $\exp(\sin(3\pi x))$, $1/(1 + 25x^2)$, $\sqrt{5}x^2$, and $\sqrt{2} \sin(2\pi x)$. Targets and checkpoints are not selected for agreement with the hypothesis. These scalings are retained because Adam's additive epsilon makes scaling part of the protocol. The saved SVD cutoff is numerical, not an exact-rank statement. Omitted energy is reported as unresolved, not automatically as nullspace.

6 What the matched measurements show

The main comparison is shown on the following landscape page so that the spectral and acquisition-time panels remain legible.

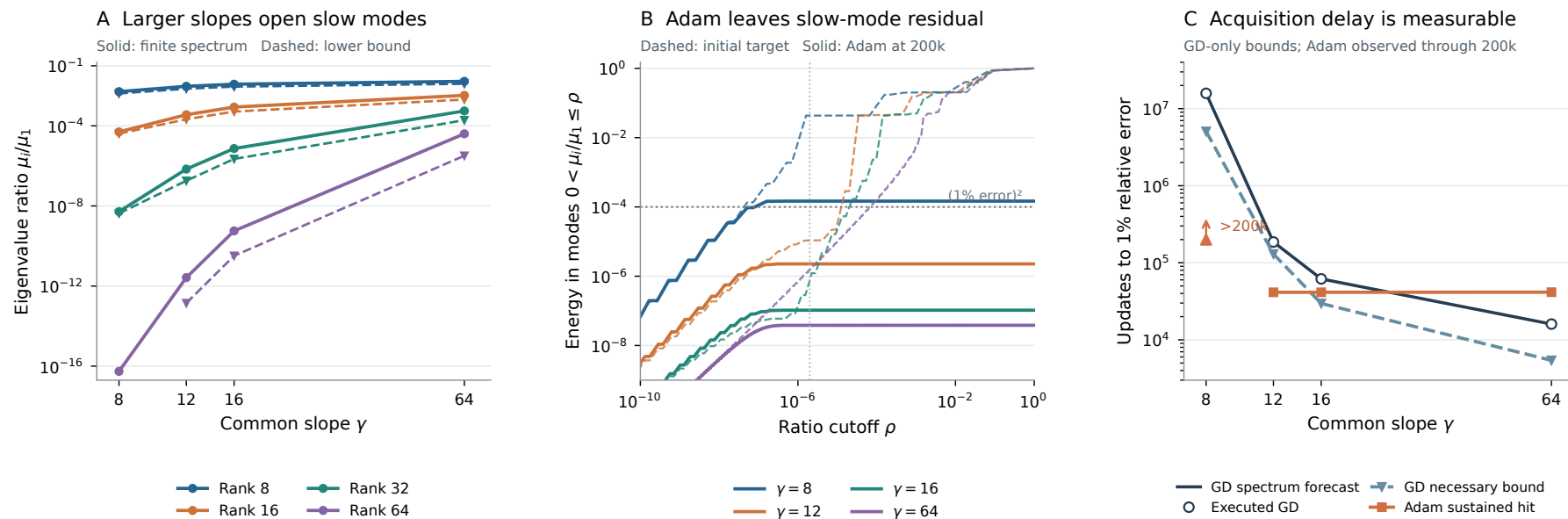


Figure 1: Gamma, needed directions, and acquisition. A: finite-kernel ratios and numerical cross-gamma theorem bounds, referenced to gamma 8. B: initial target energy and common-recipe Adam residual energy after 200,000 updates, accumulated over resolved positive modes. Energy 10^{-4} corresponds to 1% relative error. C: executed GD crossings, full-spectrum forecasts, compact necessary GD times, and sustained Adam acquisition through 200,000 updates. GD has longer executed trajectories; the Adam arrow denotes its own budget. Target and geometry match across panels. Spectral and timing bounds are different statements; no theoretical time curve is asserted for Adam.

The spectral theorem is informative, with measurable slack

At rank 32 the old ratio is 5.2174×10^{-9} . Bounds at gammas 12, 16, and 64 are 1.8172×10^{-7} , 2.2340×10^{-6} , and 1.9195×10^{-4} , retaining about 26%, 30%, and 35% of actual new ratios. The bound is within a factor of four at this prescribed rank while exposing a large explicit gamma effect. Rank 64 is less tight: bounds retain only 5.3%, 5.7%, and 8.0%. The gamma-8 baseline rank-64 bound is numerically unresolved and labeled accordingly. Ranks are not selected after seeing tightness.

The slow subspace explains the remaining gamma-8 error

At illustrative cutoff $\rho = 2 \times 10^{-6}$, gamma 8 has initial positive slow energy 0.0433333. After 200,000 common-recipe Adam updates, the remaining energy is 1.4709443×10^{-4} , alone exceeding the 10^{-4} budget for 1% error. Total relative error is 0.0121283; 99.9998% of residual energy lies in that band. Adam removed about 99.66% of its initial energy, but the remainder prevents acquisition.

At the same cutoff, final slow residual energies are 2.2660×10^{-6} , 1.0437×10^{-7} , and 3.8307×10^{-8} at gammas 12, 16, and 64, below tolerance. The cutoff is a descriptive probe inherited from the earlier $\eta\mu \leq 10^{-6}$ analysis under $\eta\mu_1 \approx 1/2$. Main cumulative curves expose full cutoff dependence. The subspaces share a spectral definition across gamma, not the same eigenvectors.

Initial slow energy is not monotone: it is 2.85×10^{-7} at gamma 16 and 1.48×10^{-6} at gamma 64. Adam can increase a diagnostic subspace's residual energy: at 20,000 updates the gamma-12 and gamma-16 slow energies exceed their initial values. These are projected residuals, not necessarily untouched initial target content.

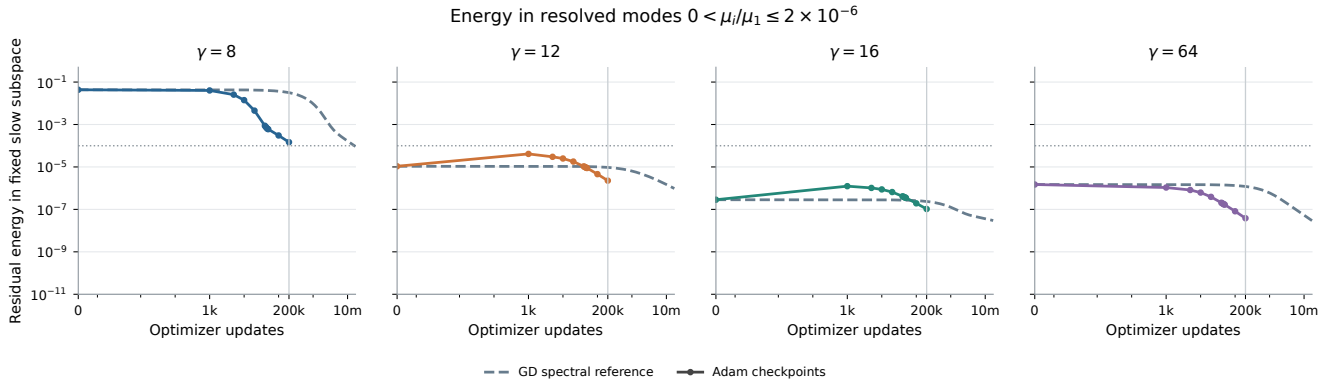


Figure 2: Evolution within the slow band. The cutoff is 2×10^{-6} for every gamma. GD curves follow its target-weighted spectrum; Adam values are saved readout checkpoints projected onto each finite kernel's modes. Energies use the original target norm. Lines between Adam checkpoints do not determine exact acquisition time; every-update traces do.

Sustained acquisition reveals the optimizer schedule

Common Adam first crosses 1% at 4,084, 1,361, and 333 updates for gammas 12, 16, and 64. Sustained crossings are 41,397, 41,412, and 41,539. Intervening oscillations and rate decay prevent treating first crossings as settled acquisition. Gamma 8 never crosses within 200,000 updates. Its validation-selected recipe also does not cross; selected gamma 64 reaches sustained acquisition at 9,917. Recipe sensitivity matters.

Executed GD reaches 1% at 15,798,313, 186,057, 61,792, and 16,013 updates. Compact necessary times are 5,072,048, 130,057, 29,640, and 5,421. The full constructed-spectrum forecast agrees with executed crossings; conservative bounds retain about 32%, 70%, 48%, and 34% of observed delay. These times do not follow by substituting panel A's ratio lower bounds.

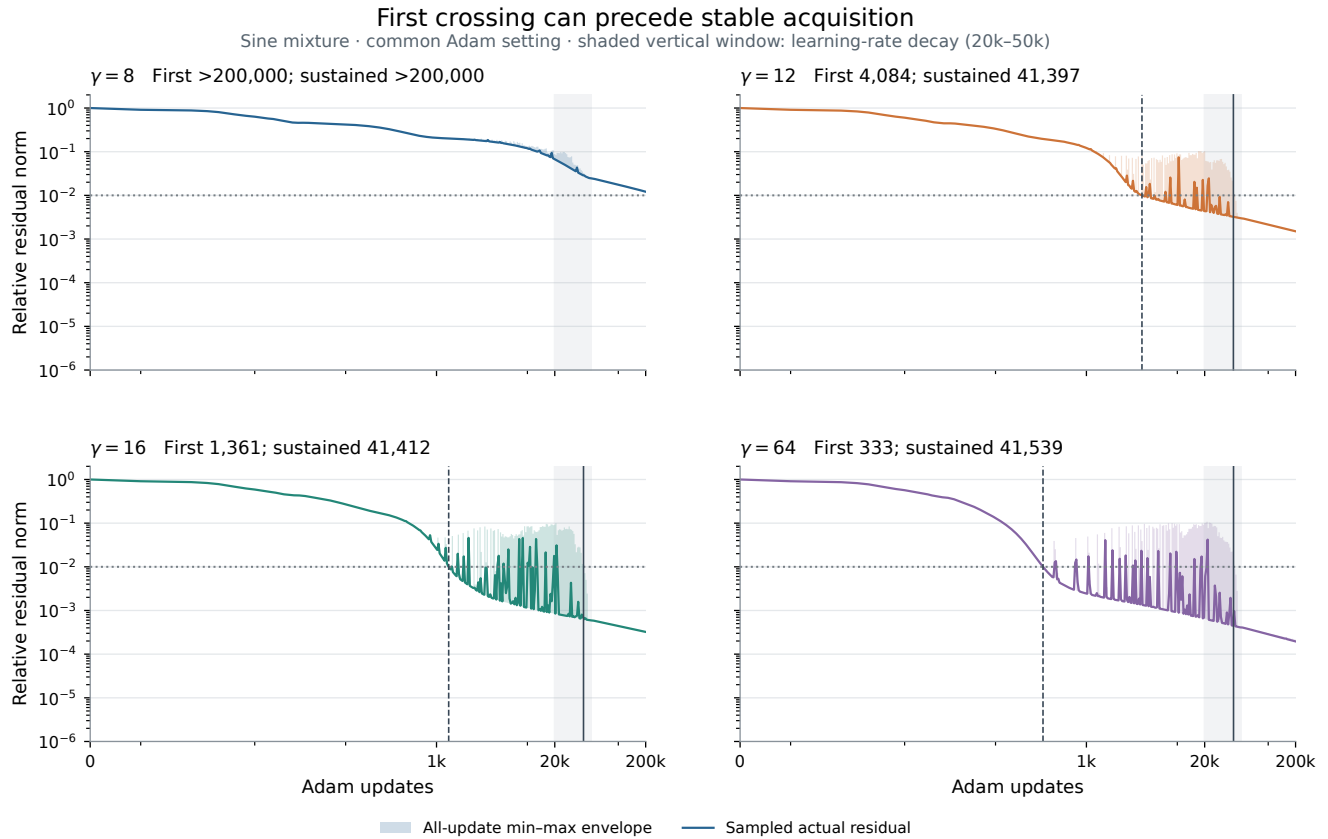


Figure 3: First crossing is not sustained acquisition. Actual per-update errors are summarized by labeled minima and maxima within update bins, without EMA smoothing. The schedule is the archived common recipe. Crossings are recomputed from every update, not these visual summaries. Rate decay explains how substantially different first crossings coexist with similar sustained crossings.

Five-target sensitivity prevents a universal claim. In selected Runge runs at gammas 12 and 16, the first 1% crossing is update 122; last recrossings postpone sustained acquisition to 199,808 and 199,710. Only 193 and 291 confirming states remain through the endpoint. Common-recipe single-sine final error also does not improve monotonically with gamma. These exceptions are retained.

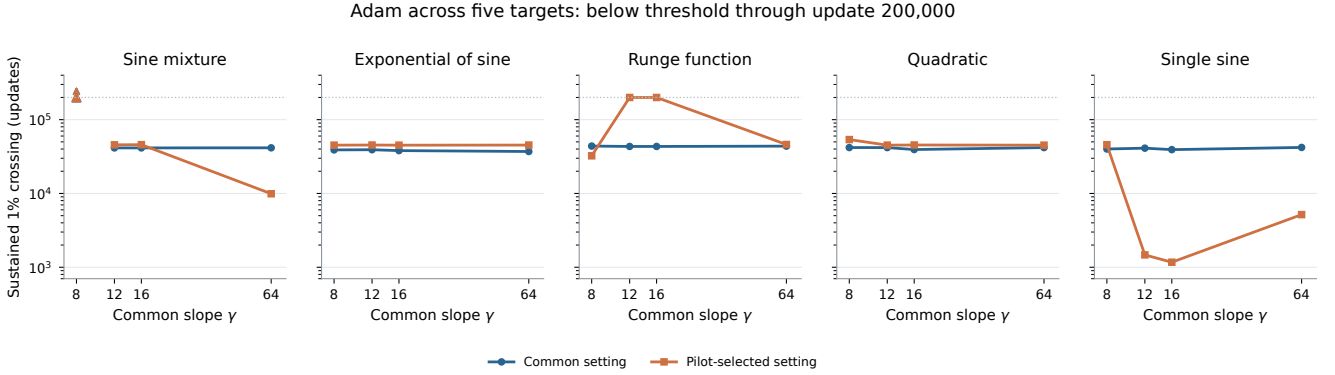


Figure 4: Target and recipe sensitivity. Every archived target is included with the same horizon-qualified definition. Recipes were selected by original late-window validation error, not fastest acquisition or agreement with this kernel argument. Arrows denote non-acquisition within 200,000 updates.

7 Numerical verification and reproduction

Adam analysis covers 40 cases and 12 saved checkpoints per case. Direct feature reconstruction matches archived hashes. All 480 residual norms agree with saved measurements within 9.14×10^{-16} . The most negative total-minus-projected energy is -2.89×10^{-15} , consistent with floating-point subtraction; this signed discrepancy is retained. Omitted-SVD energy is monitored, not labeled irreducible error.

A fresh rectangular SVD and direct sample-space residual projection reproduce band energies across all 480 checkpoint/target/view cases at seven ratio cutoffs. The largest band-energy discrepancy is 5.02×10^{-14} and energy closure differs by at most 5.33×10^{-14} . At the displayed cutoff, common-recipe discrepancies are below 3.54×10^{-16} . At gamma 16 the recomputed numerical retained rank is larger by three, and at gamma 64 by one. These are tiny-cutoff numerical rank changes, not a change in the true feature space, and do not affect the 2×10^{-6} band. The gamma-8 obstruction persists at the stricter cutoff 2×10^{-7} , where remaining energy is 1.45561×10^{-4} .

A fresh read-only persistent-archive audit processed 400 trace chunks and all states 0 through 200,000 for the 40 selected series. First, last-above, and sustained crossings agree exactly with both remote and local arrays at all six recorded tolerances. Remote metadata and local compact-manifest hashes agree. This used one CPU for 30 seconds and no GPU. Only requested scalar errors were retrieved; old data was unchanged.

The explicit Toeplitz increment agrees with independently computed direct rows within 5×10^{-16} . The small-matrix calculation checks positivity resolution, old trial-space gaps, relative correction conditioning, and two whitening formulations. These are floating-point checks, not outward-rounded certificates for new ratio bounds. The implemented C_i uses $U_i^T K_{\gamma_a} U_i$ instead of the equivalent eigenvalue diagonal to preserve the measured trial-space identity.

All 13 new focused tests pass. The full non-slow suite reports 823 passed, 17 failed, 9 skipped, and 4 deselected; the 17 failure identifiers exactly match the pre-existing baseline.

The [gamma_optimizer_access_evidence](#) contains source hashes, array conventions, unresolved statuses, retained ranks, and every plotted quantity. Full scalar traces remain in the persistent archive; compact envelopes and the independent audit accompany this study. Reproduce locally with one BLAS thread:

```
export OPENBLAS_NUM_THREADS=1 VECLIB_MAXIMUM_THREADS=1
python -m experiments.expD36_frozen_gamma_probe.gamma_ratio_bound
python -m experiments.expD36_frozen_gamma_probe.adam_spectral_analysis
python -m experiments.expD36_frozen_gamma_probe.adam_projection_audit
python -m experiments.expD36_frozen_gamma_probe.optimizer_access_figure
latexmk -pdf -outdir=/tmp/gamma-optimizer-latex \
docs/gamma_optimizer_access_note.tex
```

The optional archive audit is in `adam_trace_audit.py`: `extract` reads the persistent run directory; `verify` compares the scalar archive with local compact evidence. No least-squares refit, spectral solve, or detached diagnostic changes a trained readout. No new training was performed.